BASES AND PLACE VALUE ON ONE PAGE

Each digit counts a power of the base

Binary, ternary and decimal share one rule, which reappears in standard form and in the states of qubits and qutrits.

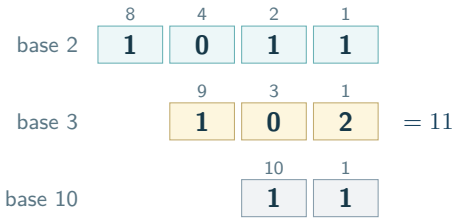
THE RULE

A numeral is a sum of digits times powers

In base b the digits run from 0 to $b-1$, and the digit in position k counts copies of b^k:

$$(d_n \ldots d_1 d_0 \,.\, d_{-1} d_{-2} \ldots)_b \quad = \quad \sum_k d_k \, b^k$$

so eleven is written $1011_2 = 8+2+1$ in binary, $102_3 = 9+0+2$ in ternary and $11_{10} = 10+1$ in decimal. Positions after the point carry negative powers, which is why $\frac{1}{3}$ is the tidy 0.1_3 in ternary while it repeats as $0.333\ldots_{10}$ and $0.0101\ldots_2$.



Eleven in three bases; the small numbers are the place values b^k.

CONVERTING

Divide by the base, keep the remainders

Each division by b peels off the last digit, so reading the remainders from the bottom up gives the numeral:

$$\begin{aligned} 11 \div 3 &= 3 \text{ r } \mathbf{2} \quad \uparrow \\ 3 \div 3 &= 1 \text{ r } \mathbf{0} \qquad \Longrightarrow \qquad 11 = 102_3. \\ 1 \div 3 &= 0 \text{ r } \mathbf{1} \end{aligned}$$

COUNTING

What n digits can hold

With n digits in base b there are b^n numerals, from 0 to $b^n - 1$, and N needs $\lfloor \log_b N \rfloor + 1$ digits. One ternary digit, a **trit**, carries $\log_2 3 \approx 1.585$ bits. Charging b for each digit, writing every number up to a million costs about 40 in binary, 60 in decimal and 37.7 in ternary, the cheapest whole base.

STANDARD FORM

Slide the point until one digit sits in front of it

Standard form records where the point belongs as a power of the base, writing $N = m \times b^e$ with $1 \le m < b$, where the exponent $e = \lfloor \log_b N \rfloor$ is one less than the number of digits:

$$3200 = 3.2 \times 10^3, \qquad 102_3 = 1.02_3 \times 3^2, \qquad 1011_2 = 1.011_2 \times 2^3.$$

Computers store floating-point numbers in exactly this binary standard form, keeping a sign, an exponent and the digits of m; since the leading binary digit of m is always 1, it is left implicit, and a 64-bit double keeps 52 further bits after it. Fractions with a factor other than 2 in the denominator never terminate in binary, so $0.1 = 0.000110011\ldots_2$ is stored approximately and Python reports `0.1 + 0.2 == 0.3` as `False`.

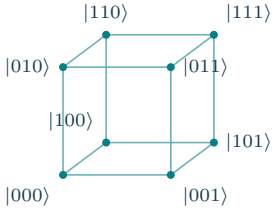
QUBITS AND QUTRITS

Place value labels the basis states

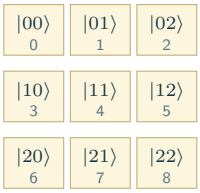
A classical bit holds 0 or 1; a qubit holds a superposition $\alpha|0\rangle + \beta|1\rangle$, and a qutrit uses the three ternary digits, $\alpha|0\rangle + \beta|1\rangle + \gamma|2\rangle$, with complex amplitudes whose squared sizes add to 1. A register of n qudits of dimension d has one basis state for every n-digit numeral in base d:

n qubits: 2^n states $|0\ldots0\rangle, \ldots, |1\ldots1\rangle$
n qutrits: 3^n states $|0\ldots0\rangle, \ldots, |2\ldots2\rangle$

so three qubits label the eight numbers 0 to 7 in binary and two qutrits label the nine numbers 0 to 8 in ternary. A general state needs d^n amplitudes to describe it, yet measuring it returns a single numeral, so n qudits deliver at most $n \log_2 d$ bits, the same capacity as n classical digits in base d.



3 qubits: $2^3 = 8$



2 qutrits: $3^2 = 9$

Conventions. A subscript names the base, so 102_3 is read "one zero two in base three"; an unmarked numeral is decimal. The ket $|x\rangle$ names the basis state labelled by the numeral x, a qudit is a quantum digit with d levels, and $\log_2 d$ converts one base-d digit into bits.

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